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Modeling Hospital Visitation Flows Through Distance-Attractiveness Tradeoffs: Integrating Human Mobility, Hospital Attributes, and Socioeconomic Context

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ABSTRACT

Hospital visitation reflects a complex destination-choice process shaped by travel cost, perceived hospital attractiveness, and neighborhood socioeconomic conditions. However, less is known about how these factors jointly structure realized hospital destination choice and how their relationships vary across travel distances and community contexts. This study integrates 4 years (2020–2023) of anonymized mobility data in Houston, TX, USA, with hospital attributes, Google Maps ratings and review counts, and 2023 American Community Survey socioeconomic indicators, including health-insurance coverage. Five predictive models were evaluated across multiple test-set proportions using 10 repeated train-test splits. The heterogeneous graph neural network (HGNN) consistently achieved the strongest predictive performance, while interpretation of the HGNN identified drive time as the dominant predictor and showed that hospital capacity, occupancy, spatial context, online visibility, ratings, and community characteristics jointly modify visitation patterns. Controlled scenario analyses indicated that rating-related distance-decay crossovers vary across socioeconomic contexts rather than following a uniform threshold. These findings highlight hospital visitation as a heterogeneous travel behavior shaped by interacting spatial, institutional, and socioeconomic factors.

1 | Introduction

Hospital visitation reflects a complex destination-choice process shaped by travel cost, hospital quality, and neighborhood socioeconomic conditions (Chen et al. 2023; McMaughan et al. 2020). Unlike many routine daily trips, healthcare-related travel decisions are shaped by healthcare needs, perceived service quality, institutional trust, and individual preferences, which may lead individuals to trade off spatial convenience against access to preferred healthcare providers. While some patients prioritize nearby hospitals to minimize travel burden, others are willing to travel longer distances to access facilities perceived to provide better services or stronger reputations (Guagliardo 2004). These tradeoffs may further vary across socioeconomic and demographic groups due to differences in mobility resources, healthcare needs, and perceptions of healthcare quality. Understanding how these interacting factors shape healthcare-related mobility can provide empirical evidence on hospital destination-choice behavior and help identify how travel burden and destination attractiveness vary across community contexts.

Multiple determinants influencing hospital visitation and healthcare-related travel behavior have been identified and empirically examined. Distance decay is widely incorporated into measures of potential spatial accessibility, such as the Enhanced Two-Step Floating Catchment Area (E2SFCA) method, in which the influence of healthcare providers is weighted according to travel-time zones (Luo and Qi 2009). At the same

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time, hospital-related factors such as service quality, reputation, waiting time, hospital capacity, and institutional level also substantially influence healthcare destination choices (Tao et al. 2020). Existing evidence suggests that patients do not necessarily choose the nearest hospital. Instead, hospital quality, specialization, reputation, and waiting time may partially offset travel costs, particularly for specialized or long-distance healthcare trips (Moscelli et al. 2016; Aggarwal et al. 2024). Recent large-scale mobility evidence further indicates that hospital bypass behavior reflects a trade-off between spatial proximity and hospital attributes and that this trade-off varies across socioeconomic groups (Xia et al. 2026). Population socioeconomic characteristics, including race, income, age, and educational attainment, may therefore shape healthcare travel decisions through differences in mobility resources, healthcare needs, and destination preferences.

Previous research has demonstrated that healthcare utilization is jointly associated with spatial accessibility, hospital characteristics, and population socioeconomic conditions. Spatial accessibility studies commonly incorporate healthcare supply indicators, such as provider capacity, together with population demand and travel impedance. Recent studies have further shown that healthcare-accessibility estimates and their interpretation can vary across spatial aggregation scales and demographic grouping schemes (Tao et al. 2025; Chen et al. 2026), while medical studies have examined the combined associations of hospital quality, travel cost, and socioeconomic characteristics with healthcare utilization. However, these studies have primarily focused on measuring potential access or estimating the average associations between these factors and healthcare use. Comparatively less attention has been given to how travel burden changes the relative importance of hospital attributes, how these distance-attribute relationships vary across community contexts, and how they jointly structure observed flows from residential origins to competing hospitals. Addressing these questions is important for understanding hospital visitation as a realized destination-choice behavior rather than solely as a condition of potential spatial accessibility.

Understanding the compounding effects of hospital and community attributes requires robust, real-world data on healthcare-related travel behavior. Advances in location tracking technologies have enabled detailed analysis of human mobility and hospital visitation patterns. For example, the Advan Research Foot Traffic/Weekly Patterns Plus dataset, accessed through Dewey Data, provides aggregated and anonymized origin–destination observations derived from mobile devices (Advan Research 2025). These flows link residential neighborhoods to hospitals and support analysis of revealed healthcare-related mobility and destination choice (Wang et al. 2021). Google Maps ratings and review counts provide complementary indicators of perceived hospital attractiveness, online reputation, and public visibility. Integrating these sources enables a behavior-oriented analysis of hospital visitation dynamics.

Mobility flow modeling has become an important area of research for understanding human movement and destination-choice behavior. Various methods, including multilayer perceptrons (MLPs) (Zhang et al. 2024), deep gravity models (Simini et al. 2021), and heterogeneous graph neural networks (HGNNs) (Terroso Sáenz et al. 2023), have been developed to model complex mobility flows and destination-choice behavior. These approaches have been applied across diverse contexts, including museum visitation, tourism mobility, transportation flows, and inter-city travel behavior. However, their relative effectiveness in modeling healthcare-related mobility remains uncertain, particularly when accounting for the combined influences of travel cost, hospital quality, and population socioeconomic characteristics. Moreover, while recent machine learning approaches improve predictive capability, less attention has been paid to how these models capture heterogeneous healthcare travel decisions and distance-quality tradeoffs across different population groups. Addressing these questions is important for advancing the understanding of healthcare-related mobility and hospital choice behavior.

Building on previous research that links spatial accessibility, hospital characteristics, and socioeconomic conditions to healthcare utilization, this study analyzes 4 years (2020–2023) of Advan Research mobility data accessed through Dewey Data in Houston, TX, USA. The analysis incorporates hospital capacity, occupancy, Google Maps ratings and review counts, community socioeconomic characteristics, and observed Drive Time between CBGs and hospitals. Rather than seeking only to confirm whether these factors are associated with healthcare utilization, the study examines how they jointly structure observed neighborhood-to-hospital visitation flows and how their relationships change across travel distances and community contexts. Specifically, this study aims to: (1) integrate social sensing data, hospital attributes, community characteristics, and observed mobility flows within a hospital destination-choice framework; (2) compare five predictive models for healthcare-related mobility; and (3) identify distance-dependent hospital-attribute patterns and examine whether the Rating-Drive Time relationship varies across community socioeconomic contexts. The contribution lies in modeling realized hospital visitation within a competing origin–destination system and examining nonlinear, distance-dependent relationships that are not directly represented by conventional accessibility measures or average-effect utilization models.

2 | Related Work

2.1 | Factors Influencing Healthcare-Related Travel Behavior

Understanding the factors influencing hospital visitation and healthcare-related travel behavior is important for explaining how individuals make healthcare destination-choice decisions. A well-established theoretical framework for understanding healthcare-seeking behavior is Andersen's Behavioral Model of Health Services Use. Originally developed by Ronald M. Andersen in 1968 to explain family use of medical services, the model proposed that healthcare utilization is shaped by three key dimensions: predisposing factors (e.g., age, gender, education, social structure, and health beliefs), enabling factors (e.g., income, insurance coverage, and spatial accessibility), and need factors (both perceived and evaluated health status) (R. Andersen 1974). The model has since undergone several major revisions. The second generation

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(Andersen and Aday 1978) expanded the framework from the individual to the societal level, integrating system-level accessibility conditions. The third version (R. M. Andersen 1995) introduced feedback mechanisms among health behaviors, outcomes, and healthcare systems, establishing a dynamic and multidimensional structure. Subsequent developments of the model further incorporated contextual influences such as policy environment, community characteristics, and healthcare infrastructure, allowing applications across diverse populations and spatial scales (Alkhawaldeh et al. 2023; Babitsch et al. 2012). This conceptual framework provides an important foundation for understanding hospital visitation as a healthcare-related travel decision shaped by interacting behavioral, socioeconomic, and spatial factors.

Building on this framework, previous studies have identified two major groups of determinants influencing healthcare-related travel behavior and hospital choice. Hospital-related factors, such as medical quality, reputation, patient experience, insurance coverage, hospital size, and proximity, represent institutional conditions that shape healthcare destination choices. The relative importance of these factors varies substantially across healthcare systems and geographical contexts (Du and Zhao 2022; Wang et al. 2022, 2018). In Europe, where healthcare systems are predominantly public and geographically regulated, spatial proximity and service quality often emerge as primary determinants of hospital choice. In the Netherlands, patients with vascular diseases prioritize hospitals known for reputable cardiology departments and lower readmission rates (Varkevisser et al. 2012). Dutch breast cancer and cataract patients also demonstrate a willingness to trade travel distance for perceived service quality, although shorter travel distances generally remain preferred (Salampessy et al. 2022). In the United Kingdom, an analysis of more than 200,000 inpatient admissions across six hospitals in Derbyshire revealed that travel distance was a stronger determinant of hospital choice than hospital size, waiting time, cleanliness, or facility availability (Smith et al. 2018). In China, where healthcare systems exhibit strong institutional hierarchies, insurance reimbursement and hospital level substantially influence patient preferences. A survey conducted in Shenzhen showed that patients prioritized insurance reimbursement, hospital level, distance, cost, and waiting time when selecting healthcare facilities (Zhao et al. 2022). In the United States, where healthcare systems are characterized by fragmented insurance structures and pronounced socioeconomic disparities, race, insurance type, and household income significantly influence healthcare travel distance and hospital choice behavior (Orringer et al. 2022).

Beyond hospital attributes, demographic and socioeconomic characteristics further shape healthcare-related mobility and hospital visitation behavior. A global survey spanning 16 countries and 14 medical specialties found that individuals with higher socioeconomic status (SES) tend to prioritize service quality and provider attitude over logistical considerations such as distance or cost, whereas lower-SES populations often emphasize convenience and travel burden due to financial and transportation constraints (Lin et al. 2024). Cultural and institutional contexts also influence healthcare destination choices. In China, household registration (hukou) status significantly affects preferences for public versus private healthcare facilities, with urban residents showing stronger preferences for public hospitals than rural migrants (Tang et al. 2016). Other studies have also demonstrated the effects of race, age, gender, and educational attainment on healthcare travel decisions and hospital preferences (Mhlanga and Hassan 2022; Ohlson 2020; Yu et al. 2017). These findings suggest that healthcare visitation is shaped not only by spatial accessibility but also by heterogeneous travel preferences and socioeconomic constraints across population groups.

Collectively, previous studies demonstrate that healthcare utilization is shaped by the combined influence of spatial accessibility, hospital capacity and quality, and socioeconomic conditions. Accessibility models have made important advances by incorporating healthcare supply and population demand, while medical utilization studies have documented relationships among travel cost, hospital characteristics, socioeconomic status, and healthcare use. Nevertheless, potential accessibility and observed hospital destination choice remain conceptually distinct. The availability of healthcare resources within a geographic area does not necessarily reveal which hospital residents actually visit, particularly when multiple hospitals compete for patients and individuals may bypass nearby facilities. In addition, models estimating overall or average associations may not reveal how the relative importance of hospital attributes changes with travel time or whether these distance-dependent relationships vary across community socioeconomic contexts. These issues motivate a behavior-oriented analysis of observed neighborhood-to-hospital visitation flows.

2.2 | Models for Mobility Flow Prediction

Mobility flow prediction plays an important role in understanding human movement and destination-choice behavior across complex spatial systems. The increasing availability of large-scale human mobility datasets has facilitated the development of data-driven predictive models. Traditional spatial interaction approaches, such as gravity models inspired by Newton's law of gravitation, estimate flows based on population size and distance relationships (Zipf 1946). While effective in capturing general distance-decay patterns, these models often assume globally constant relationships and therefore have limited ability to represent behavioral nonlinearities and spatial heterogeneity in real-world mobility systems.

Machine learning approaches provide greater flexibility for modeling heterogeneous mobility behavior and complex spatial interactions. Techniques such as Gradient Boosting can iteratively learn nonlinear relationships from empirical data without relying on predefined functional assumptions (Guo et al. 2025). Deep learning methods further advance mobility modeling by capturing complex interactions among geographic, socioeconomic, and mobility-related variables. Among these approaches, Deep Gravity (Simini et al. 2021) integrates gravity-law principles within a neural network framework, enabling end-to-end learning of spatial interaction

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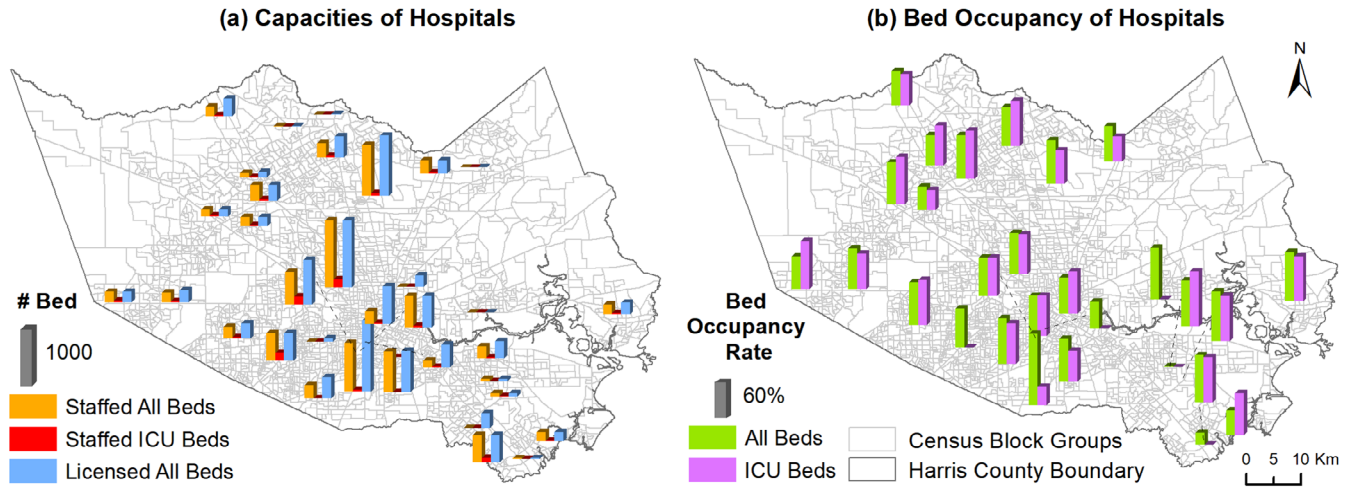


FIGURE 1 | Spatial distribution of hospital capacities (a) and bed occupancy rates (b) (Gray bars serve as unified visual references for comparability across variables).

patterns while preserving interpretability. Subsequent studies have applied Deep Gravity models to diverse mobility contexts, including freight transportation, intercity movement, and regional population flows (Cabanas-Tirapu et al. 2025; Yang et al. 2023; Zhao et al. 2025).

Graph Neural Networks (GNNs) further extend mobility flow modeling by leveraging graph structures to capture spatial dependencies and complex interactions within mobility systems. Existing studies have combined GNNs with temporal models to analyze dynamic mobility flows and travel behavior (Wang et al. 2024). Building upon this framework, Heterogeneous Graph Neural Networks (HGNNs) can model multiple node and relationship types simultaneously, enabling the integration of spatial, functional, and semantic relationships within mobility networks (Liu and Meidani 2024; Liu et al. 2025; Xie et al. 2024). These approaches have demonstrated strong capabilities in capturing heterogeneous mobility patterns across complex transportation systems.

Despite these methodological advances, applications of machine learning and graph-based mobility models to healthcare-related travel behavior remain relatively limited. Most existing mobility prediction studies focus on commuting, tourism, freight transportation, or general traffic flows rather than healthcare-related mobility. Healthcare visitation differs from many routine mobility contexts because destination choices are simultaneously shaped by travel cost, perceived hospital quality, institutional characteristics, and socioeconomic constraints. Moreover, the relative effectiveness of Deep Gravity and HGNN approaches in modeling hospital visitation flows remains unclear. Deep Gravity represents nonlinear origin–destination interactions, whereas HGNN additionally represents CBGs and hospitals as distinct node types connected through observed visitation relationships. Comparing these approaches therefore provides evidence on whether heterogeneous graph structure and training-edge message passing add predictive information beyond conventional gravity-inspired interactions.

3 | Data and Method

3.1 | Data Collection and Preprocessing

This study focuses on hospital visitation behavior within Harris County, TX, USA, which includes the City of Houston and surrounding suburban areas. All analyses were conducted within Harris County because healthcare facilities and patient catchment areas extend beyond the administrative boundary of the city, forming an interconnected healthcare mobility system.

3.1.1 | Healthcare Capacity and Occupancy Data

Hospital capacity and occupancy data were obtained from CovidCareMap, which compiles Healthcare Cost Report Information System (HCRIS) data. We initially identified 50 general acute care hospitals in Houston and excluded specialty, rehabilitation, and psychiatric facilities because their service scopes and patient populations differ from general hospital care. Thirty-five hospitals could be matched across all required datasets and were retained for analysis. Five attributes were used: Staffed All Beds, Staffed ICU Beds, Licensed All Beds, All Bed Occupancy Rate, and ICU Bed Occupancy Rate. Figure 1 maps these attributes. Staffed All Beds ranged from 0 to 1310 (mean 281.43), Staffed ICU Beds from 0 to 162 (mean 30.23), and Licensed All Beds from 4 to 1403 (mean 368.97). All Bed Occupancy Rate ranged from 0% to 86% (mean 51.26%), and ICU Bed Occupancy Rate from 0% to 92% (mean 43.24%).

3.1.2 | Human Mobility Data

Human mobility data were obtained from the Advan Research Foot Traffic/Weekly Patterns Plus dataset through the Dewey Data platform (Advan Research 2025). The dataset provides anonymized origin–destination visits from residential Census

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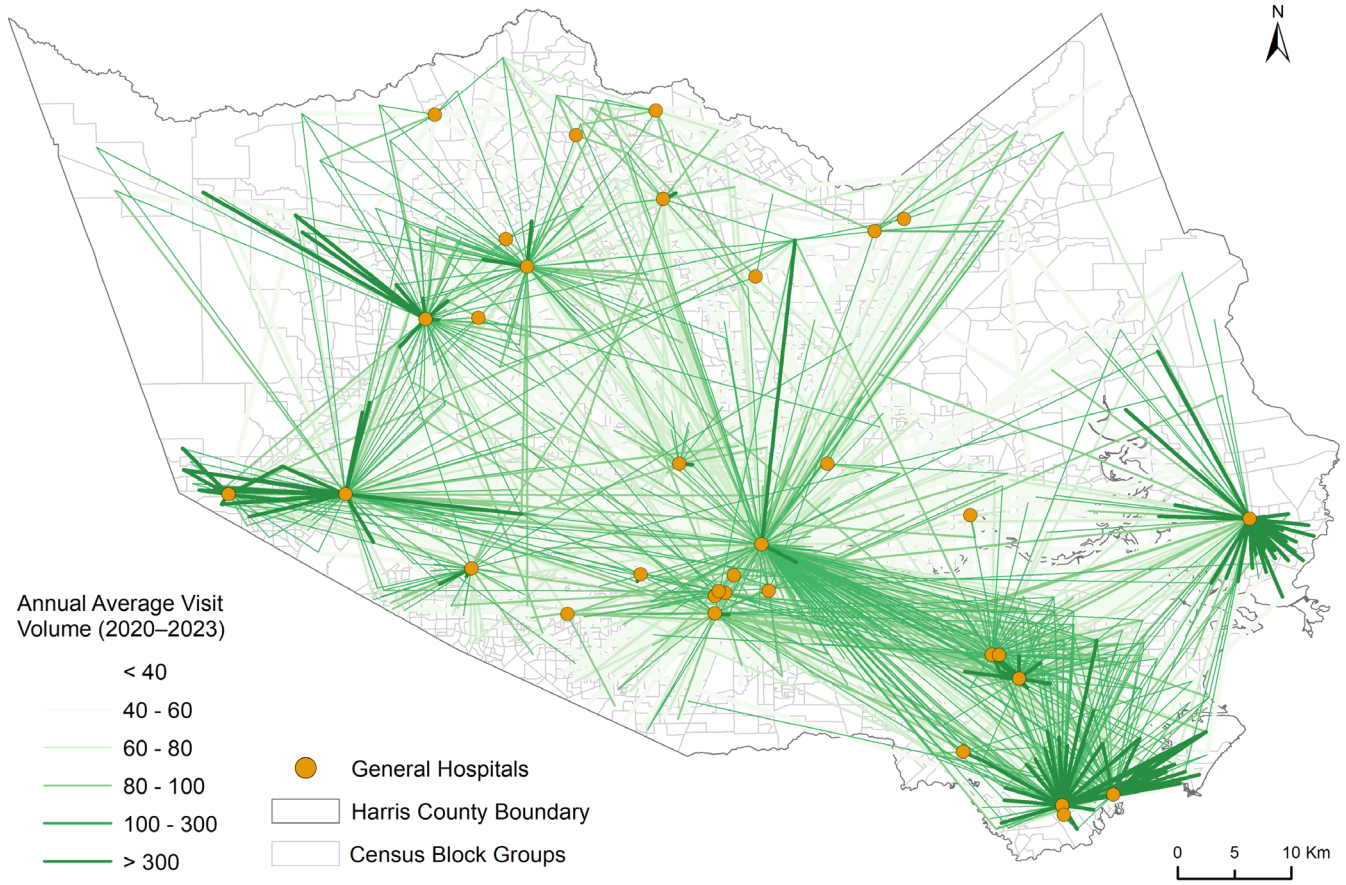


FIGURE 2 | Spatial patterns of 4-year average visits to general hospitals in Houston (2020–2023).

Block Groups (CBGs) to Points of Interest. We matched 35 general acute care hospitals in Harris County to the CovidCareMap data; these hospitals recorded 1,657,488 visits during 2020–2023.

Weekly visitation records from January 2020 to December 2023 were aggregated annually and then averaged across the 4 years. The study period spans the major COVID-19 crisis (2020–2021) and recovery period (2022–2023), when hospital visitation and operations may have differed. Because the study examines longer-term CBG-hospital spatial interactions rather than pandemic-stage changes, the 4-year average is used to represent the average spatial interaction structure across the complete study period, not conditions in a presumed normal period. The initial coverage comprised 2830 Harris County CBGs and 35 hospitals. After linking complete hospital, Drive Time, and 2023 ACS attributes, the analytical graph contained 16,783 observed flows from 1155 origin CBGs to 35 hospitals. Only observed flows were modeled; unobserved pairs were not added as zero-valued negative samples. Figure 2 shows the resulting 4-year average flows.

Several notable long-distance flows were retained in the final analytical dataset, including flows associated with Census Block Group FIPS 482019801001 near George Bush Intercontinental Airport. Such observations may reflect atypical mobility contexts, including airport-related visitors or medical travel, and should therefore be interpreted cautiously rather than as conventional residential healthcare demand.

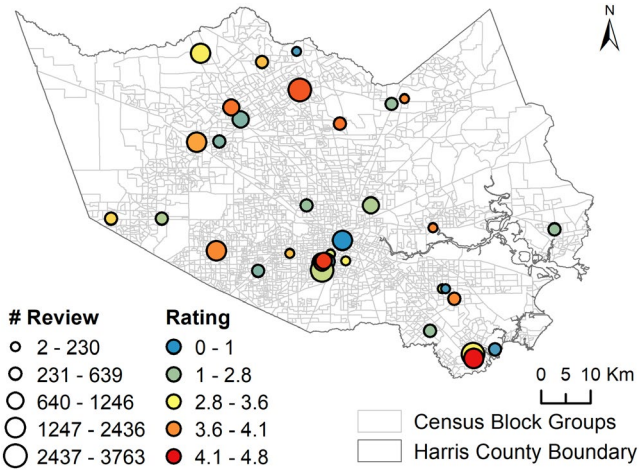


FIGURE 3 | Spatial distribution of hospital user experience indicators, where the number of reviews reflects online visibility and ratings represent perceived service quality.

Advan Research visitation data were used as a proxy for observed healthcare visitation. Because the data derive from mobile-device samples, they are subject to coverage, sampling, and representativeness limitations and are not ground-truth patient volumes. Visitation was normalized within each CBG as the percentage of its observed visits allocated among the 35 hospitals, so the observed percentages sum to 100% per CBG. Models predicted these observed CBG-hospital percentages; unobserved pairs

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were not assigned zero targets, and HGNN predictions were not constrained by a post-decoder CBG-wise softmax.

3.1.3 | Google Maps Data

Google Maps supplied two inputs collected in 2024. Hospital ratings and review counts were web-scraped for the 35 hospitals to represent contemporary reputation and service perception rather than a specific historical period. Ratings ranged from 1 to 4.8 (mean 3.45), and review counts from 2 to 3763 (mean 966.74). Drive Times from all CBGs to the hospitals were obtained through the Google Maps Distance Matrix API and ranged from 1.65 to 69.95 min (mean 27.37 min). Figure 3 maps hospital ratings and review counts.

Google Maps ratings and reviews may contain demographic and behavioral bias because contributors are more likely to be digitally active. Accordingly, they are interpreted as indicators of perceived service quality, online reputation, visibility, and public awareness rather than objective clinical quality. Review volume may also reflect accumulated visitation and should not be treated as an independent measure of hospital performance.

3.1.4 | Socioeconomic Data

CBG socioeconomic characteristics were obtained from the 2023 American Community Survey 5-Year estimates. The indicators were total population; percentages under age 18 years old and age 65 years old or older; percentages of Hispanic, non-Hispanic White, non-Hispanic Black, and non-Hispanic Asian populations; percentage with a bachelor's degree or higher; median household income; percentage of households with a vehicle; and percentage of the civilian noninstitutionalized population with health-insurance coverage. These variables represent established demographic, socioeconomic, and mobility-related determinants of healthcare use (Guagliardo 2004; Luo and Wang 2003; Wang and Luo 2005).

Table 1 lists nine hospital attributes and 13 CBG attributes used by the neural-network models, including two CBG coordinates and the health-insurance coverage measure. The dependent variable is visitation percentage, and Drive Time is the edge-level travel-cost measure. Hospital and CBG longitude and latitude were retained to represent broader spatial structure not captured by pairwise drive time alone, including hospital clustering, spatial competition, and the geographic organization of residential origins (Simini et al. 2021; Song 2023).

TABLE 1 | Summary of variables, categories, and data sources.

Categories	Variables	Data sources
Healthcare systems attributes	Staffed all Beds	CovidCareMap Dashboard
	Staffed ICU Beds	
	Licensed all Beds	
	All bed occupancy rate	
	ICU bed occupancy rate	
	No. of reviews	Google Maps Review Data
	Rating	
	Hospital longitude	CovidCareMap Dashboard
	Hospital latitude	
Population socioeconomic status	Total population	American Community Survey
	% population under 18	
	% population over 65	
	% Hispanic population	
	% White population	
	% Black population	
	% Asian population	
	% Bachelor's degree or higher	
	Median household income	
	% Households with a vehicle	
	% Population with health insurance coverage	
	Visitor departure longitude	US Census
	Visitor departure latitude	
Distance cost	Drive time	Google Map Data
Hospitals visitation	Visitation percentage	Advan Research via Dewey Data

3.2 | Predictive Models for Healthcare Facility Visits

3.2.1 | Model Descriptions and Experimental Settings

Five models were evaluated: Naive Regression, Gradient Boosting, Multi-Layer Perceptrons (MLP), Deep Gravity, and Heterogeneous Graph Neural Network (HGNN). All models were designed to predict visitation percentage using nine hospital attributes, 13 CBG attributes, and raw Drive Time in minutes. Naive Regression used ordinary least squares to estimate a single linear relationship between the concatenated predictors and visitation percentage and therefore served as the linear baseline. Gradient Boosting sequentially combined regression trees fitted to preceding prediction errors, enabling nonlinear relationships and interactions to be represented (Friedman 2001). The MLP passed the concatenated CBG, hospital, and Drive Time predictors through fully connected nonlinear hidden layers before producing a continuous visitation-percentage prediction (Hornik et al. 1989).

Deep Gravity learned nonlinear interactions among origin characteristics, destination characteristics, and travel cost. It projected CBG attributes, hospital attributes, and raw Drive Time into latent representations, concatenated them, and used an MLP decoder to predict visitation percentage (Figure 4). All models used the same observed-edge inputs and train-test partitions.

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The HGNN represented CBGs and hospitals as two node types connected by observed visitation edges (Figure 5a). Thirteen CBG features and nine hospital features were projected separately into 32-dimensional embeddings. One heterogeneous GraphSAGE layer propagated information over observed training edges and their reverse relations; held-out edges were excluded from message passing and used only as decoder queries. The decoder concatenated the CBG embedding, hospital embedding, and unstandardized Drive Time and processed the resulting 65-dimensional vector through layers of $65 \to 64 \to 32 \to 1$ with a nonnegative output. Visitation percentage served only as the target. The model used no negative pairs or CBG-wise sum-to-100 prediction constraint.

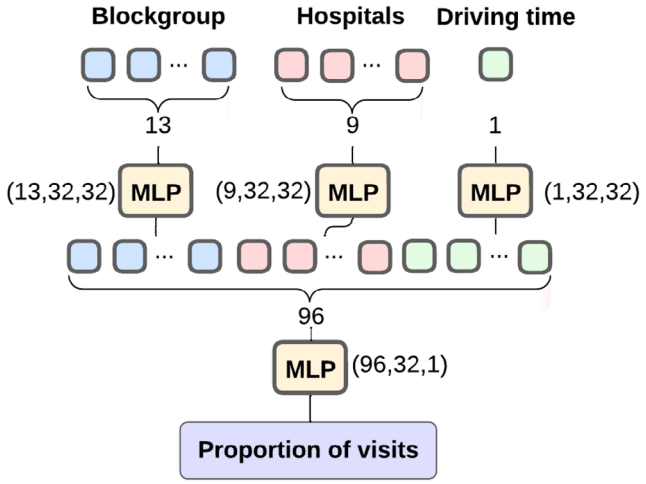


FIGURE 4 | Structure of the Deep Gravity Model.

Performance was evaluated at test-set proportions of 10%, 20%, and 30% using 10 repeated train-test splits. For each test-set proportion, the model was trained and evaluated independently across 10 repeated splits, and the held-out edges in each split were used only for final prediction and evaluation. The reported values are the mean and sample standard deviation across the 10 splits.

3.2.2 | Performance Evaluation Metrics

Performance was measured using Normalized Root Mean Square Error (NRMSE; Equation 1), Symmetric Mean Absolute Percentage Error (SMAPE; Equation 2), and Common Part of Commuters (CPC; Equation 3) (Zhang et al. 2024). NRMSE equals RMSE divided by the standard deviation of observed test values. Lower NRMSE and SMAPE indicate better performance, whereas higher CPC indicates greater overlap between observed and predicted flows. Negative predictions, where applicable, were clipped to zero for metric calculation.

$$\mathrm{NRMSE} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2}}{\sigma_y} \tag{1}$$

$$\mathrm{SMAPE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|\,y_i - \hat{y}_i\,|}{(|\hat{y}_i| + |y_i|)/2} \times 100 \tag{2}$$

$$\mathrm{CPC} = \frac{2\sum_{i=1}^{N}\min\left(y_i, \hat{y}_i\right)}{\sum_{i=1}^{N}\left(y_i + \hat{y}_i\right)} \tag{3}$$

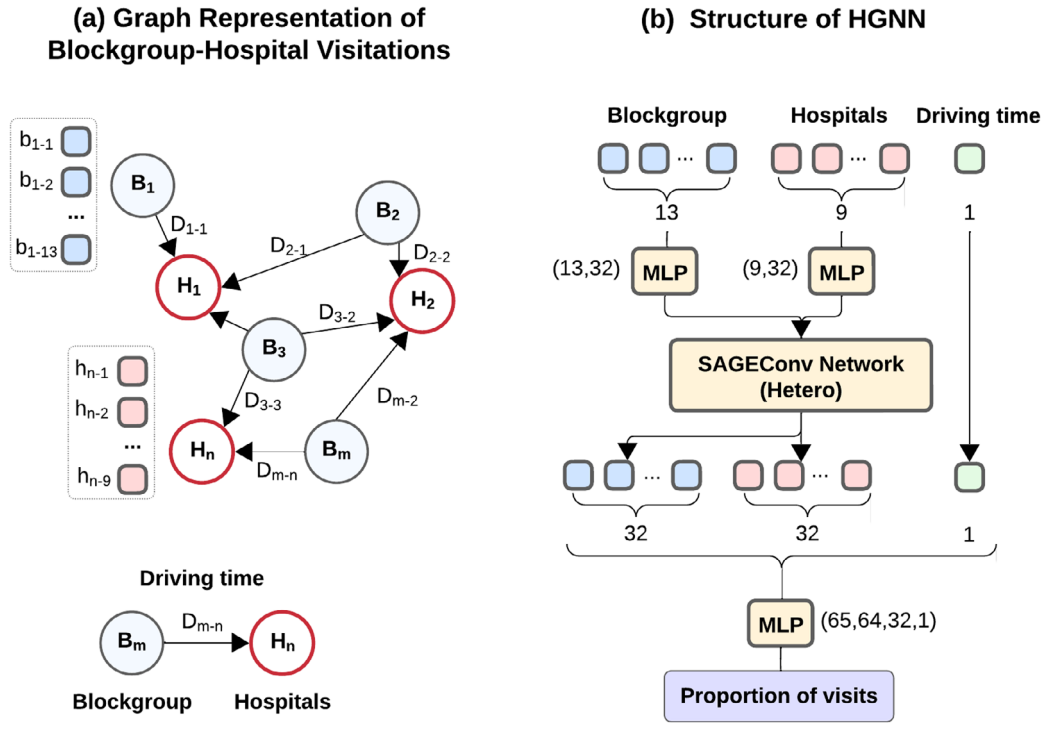


FIGURE 5 | Graphical representation and structure of the HGNN model. (a) Graphical representation of blockgroup-hospital visitations. (b) Structure of the HGNN model.

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Here, $\hat{y}_i$ and y_i are the predicted and observed values for held-out flow i, N is the number of held-out flows, and σ_y is the standard deviation of observed test values. For each test proportion, metrics were summarized across the 10 repeated splits as mean ± sample standard deviation.

3.3 | Methods of Interpretation and Scenario Analysis

The final HGNN was interpreted using SHAP (SHapley Additive exPlanations) and controlled partial-dependence-style scenarios (Friedman 2001). It was refitted on all 16,783 observed edges, so these outputs describe the fitted model rather than held-out performance. To limit unstable attribution among correlated features, Staffed All Beds, Licensed All Beds, and Staffed ICU Beds were grouped as "All Beds," while All Bed Occupancy Rate and ICU Bed Occupancy Rate were grouped as "Bed Occupancy Rate." Insurance coverage remained an independent CBG feature. Mean absolute SHAP values ranked importance, and signed values indicated the direction of contributions.

HGNN scenarios examined how hospital attributes and community context modified distance decay. For hospital attributes, the focal variable was set to its observed minimum, mean, or maximum while other nonspatial attributes were held at their means and actual coordinates were retained. Predictions used each observed CBG-hospital query and its original Drive Time and were averaged in observed 5-min strata from 0 to 70 min. The rating-by-SES analysis crossed higher and lower values of each population variable with the highest and lowest Hospital Rating, producing four curves per panel. These controlled simulations are not causal or individual-level estimates.

For each scenario, the first intersection between the highest- and lowest-rating curves was recorded; none was reported when the curves did not cross within the observed Drive Time range. Crossovers represent model-based transitions under the specified synthetic conditions, not fixed behavioral thresholds.

4 | Results

4.1 | Model Performance

Table 2 reports mean ± standard deviation across 10 repeated train-test splits at 10%, 20%, and 30% test-set proportions. HGNN achieved the strongest performance at every reported test proportion. At 10% test, HGNN obtained NRMSE = 0.6081 ± 0.0258, SMAPE = 56.0842% ± 1.5010%, and CPC = 0.7530 ± 0.0073. At 20% test, it achieved 0.6137 ± 0.0205, 56.8364 ± 1.7721%, and 0.7509 ± 0.0063, respectively; at 30% test, it achieved 0.6163 ± 0.0189, 56.1804% ± 1.9361%, and 0.7495 ± 0.0076. Deep Gravity ranked second, followed by MLP, Gradient Boosting, and Naive Regression. The consistent advantage of the flexible nonlinear models over Naive Regression, together with the nonlinear distance-decay and crossover patterns identified in the subsequent SHAP and scenario analyses, suggests that a purely linear and additive representation may not adequately represent the observed hospital visitation relationships. The small change in HGNN performance across test proportions indicates comparatively stable generalization under different amounts of held-out data.

HGNN's advantage over Deep Gravity was consistent but moderate, suggesting that both nonlinear gravity-like interactions and graph-based neighborhood information contribute to hospital visitation prediction. The HGNN's explicit separation of CBG and hospital node types, together with message passing over training-only observed edges, provided additional predictive value beyond feature concatenation alone. Because HGNN performed best across all three metrics and test proportions, subsequent SHAP and scenario analyses were conducted based on the trained HGNN model.

Figure 6 compares observed and HGNN-predicted visitation flows for one representative 20% test split. All 3356 held-out edges were mapped using common flow-percentage classes. Observed visitation percentages ranged from 0.10% to 81.52%, predicted values ranged from 0.52% to 78.33%, and the mean absolute error was 3.41 percentage points. The spatial concentration and major hospital-centered flow patterns were reproduced in the predictions, while differences remain for some high-percentage flows.

For test flows with observed visitation percentages above 40%, the HGNN underestimated the observed value by an average of 15.48 percentage points. These residuals indicate that a subset of highly concentrated destination choices may reflect hospital-specific services, institutional reputation, or atypical origin contexts not fully represented by the available attributes. Figure 6 is a visual assessment of one test split and should be interpreted together with the results from the 10 repeated train-test splits in Table 2.

4.2 | Factors Influencing Hospital Visitations

Figure 7 presents the top 20 grouped factors influencing fitted visitation percentages according to SHAP values from the HGNN. Drive Time was the most influential predictor and showed a strong distance-decay pattern: higher travel times were associated predominantly with negative contributions, whereas shorter times supported larger predicted visitation percentages. Hospital and visitor coordinates also ranked highly, indicating that broader metropolitan spatial structure contributes information beyond pairwise Drive Time. These coordinate effects should be interpreted as learned spatial context rather than causal effects of longitude or latitude themselves.

The grouped "All Beds" factor ranked second, indicating that hospital scale and resource availability substantially influenced fitted visitation. Hospital Longitude ranked third and grouped Bed Occupancy Rate ranked fourth, followed by Visitor Departure Longitude, Hospital Latitude, and Visitor Departure Latitude. The broad SHAP distributions for capacity and occupancy indicate nonlinear and context-dependent relationships rather than uniform positive or negative effects.

The number of review comments and Hospital Rating ranked eighth and ninth, respectively. Higher ratings were generally associated with positive SHAP contributions, consistent with

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TABLE 2 | Average model performance (mean ± standard deviation) across different test-set proportions based on 10 repeated train-test splits.

Model	Test size	NRMSE	SMAPE (%)	CPC
Naive Regression	10%	1.0004 ± 0.0009	86.3994 ± 1.2711	0.5704 ± 0.0069
	20%	1.0001 ± 0.0001	86.5778 ± 0.7796	0.5676 ± 0.0042
	30%	1.0001 ± 0.0001	86.3318 ± 0.5335	0.5691 ± 0.0031
Gradient Boosting	10%	0.7970 ± 0.0146	70.2281 ± 1.1224	0.6720 ± 0.0052
	20%	0.7984 ± 0.0105	70.3984 ± 0.5888	0.6697 ± 0.0039
	30%	0.7989 ± 0.0152	70.2614 ± 0.6752	0.6699 ± 0.0050
MLP	10%	0.6982 ± 0.0207	72.1134 ± 1.7095	0.7012 ± 0.0099
	20%	0.7023 ± 0.0268	72.8065 ± 1.4556	0.6979 ± 0.0117
	30%	0.7061 ± 0.0234	72.9666 ± 1.3481	0.6965 ± 0.0102
Deep Gravity	10%	0.6449 ± 0.0218	61.1060 ± 0.9660	0.7363 ± 0.0046
	20%	0.6478 ± 0.0161	61.3438 ± 1.1617	0.7350 ± 0.0048
	30%	0.6517 ± 0.0125	61.5541 ± 0.8807	0.7336 ± 0.0045
HGNN	10%	**0.6081 ± 0.0258**	**56.0842 ± 1.5010**	**0.7530 ± 0.0073**
	20%	**0.6137 ± 0.0205**	**56.8364 ± 1.7721**	**0.7509 ± 0.0063**
	30%	**0.6163 ± 0.0189**	**56.1804 ± 1.9361**	**0.7495 ± 0.0076**

Note: The bold values indicate the best-performing results for each test-set proportion and evaluation metric. Specifically, lower NRMSE and SMAPE values and higher CPC values indicate better model performance.

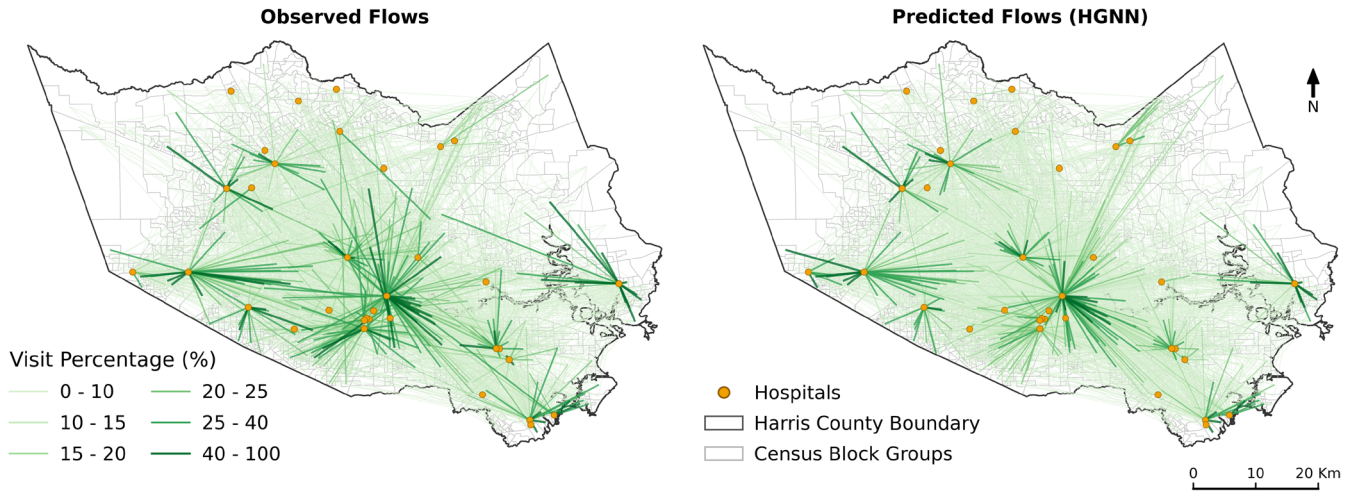


FIGURE 6 | Observed versus predicted flows by the HGNN Model on the test dataset.

ratings acting as a perceived-attractiveness signal after the model accounts for distance and spatial context. In contrast, high review counts frequently showed negative SHAP contributions. This pattern should not be interpreted as evidence that reviews discourage visitation; review volume is intertwined with hospital identity, service mix, accumulated visitation, and location, and therefore represents a conditional model association rather than an isolated causal effect.

Community characteristics contributed additional heterogeneity. Hispanic population percentage ranked highest among the demographic variables, while White population percentage, total population, bachelor's-degree attainment, Black population percentage, health-insurance coverage, income, age composition, Asian population percentage, and household vehicle availability had smaller grouped SHAP magnitudes. Health-insurance coverage therefore contributed measurable but comparatively modest incremental information after the correlated socioeconomic and spatial predictors were considered.

4.3 | Impacts of Hospital Attributes on Visitation Patterns

Figure 8 examines when and how changing hospital attributes differentiate the predicted hospital choice as travel becomes longer. The blue, yellow, and red curves represent mean, maximum, and minimum attribute scenarios, respectively, while other non-spatial attributes were held at their means and actual spatial coordinates were retained. Predictions were generated using each

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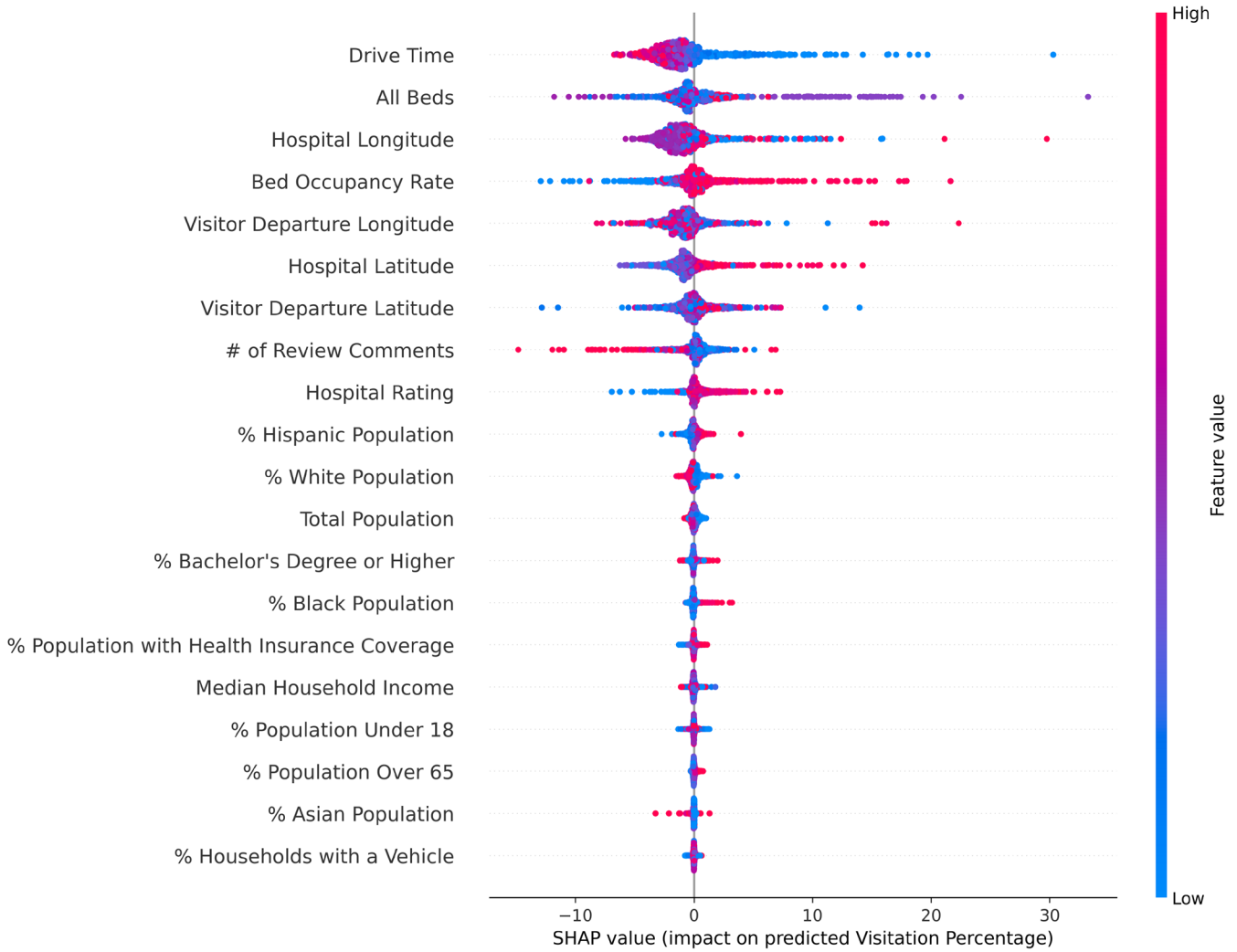


FIGURE 7 | Top 20 factors influencing hospital visitation shown by the SHAP Summary Plot of the HGNN Model.

observed edge's original Drive Time and subsequently averaged within five-minute strata. The intersections are interpreted as distance thresholds at which the relative ordering of the high- and low-attribute scenarios changes. They do not imply that an attribute has no effect before the threshold; rather, they indicate when the fitted model begins to favor the higher-value scenario over the lower-value scenario.

For All Beds, the maximum- and minimum-capacity curves first intersected at approximately 20.01 min (predicted visitation 6.35%). Before this point, greater bed capacity was not consistently associated with higher predicted visitation, whereas beyond the intersection the larger-capacity scenario became comparatively more attractive. Staffed ICU Beds showed convergent curves without a clearly defined crossover within the plotted range, suggesting no comparable distance threshold for this attribute.

All Bed Occupancy Rate displayed a different pattern. The high- and low-occupancy curves remained separated across the observed distance range and no intersection was identified. In the fitted model, this stable ordering indicates that bed occupancy remained relevant to predicted hospital choice for both nearby and longer-distance visits rather than becoming influential only after a particular travel-time threshold. ICU Bed Occupancy Rate showed a much later intersection at approximately 39.55 min (3.90% predicted visitation), outside the main range in which most visitation occurred.

The Number of Reviews scenarios intersected at approximately 10.92 min (21.31% predicted visitation). Beyond that point, the minimum-review curve was generally above the maximum-review curve. Consistent with the SHAP results, this counter-intuitive conditional pattern likely reflects correlations between review volume, hospital location, scale, and accumulated patient activity rather than a direct adverse effect of online visibility.

Hospital Rating produced a crossover at approximately 12.78 min (16.68% predicted visitation). Before this threshold, the model did not favor the maximum-rating scenario over the minimum-rating scenario. Beyond 12.78 min, however, the maximum-rating curve became higher, indicating that patients undertaking longer trips were increasingly predicted to select the higher rated hospitals. Rating therefore appears to function as a destination-attractiveness signal that becomes more consequential after patients move beyond predominantly local hospital use.

The scenarios distinguish attributes that become relevant only after a distance threshold from attributes whose fitted

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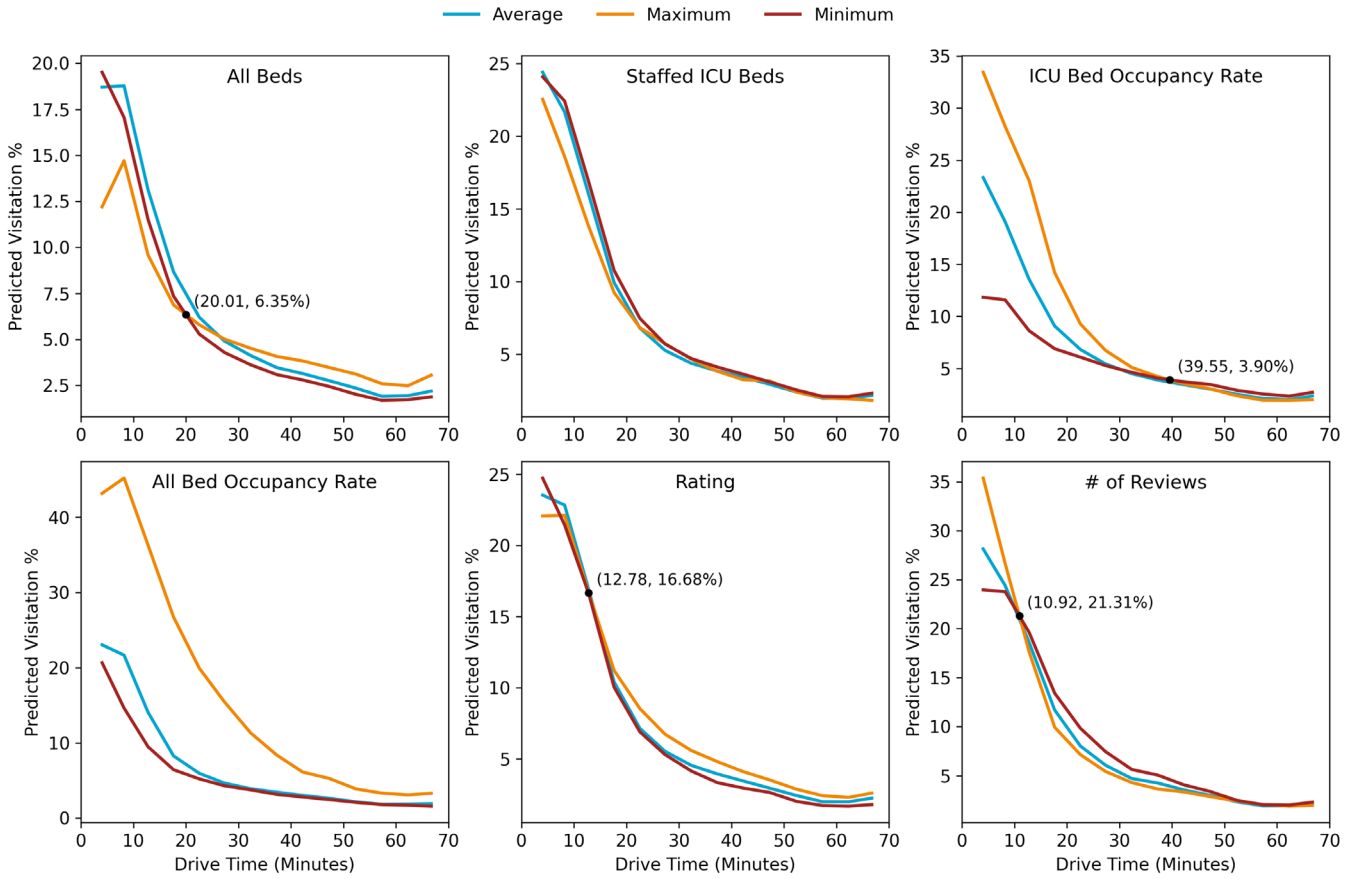


FIGURE 8 | Effects of hospital attributes on visitation distance decay patterns.

importance persists across distances. Rating and bed capacity exhibited identifiable transitions, whereas All Bed Occupancy Rate maintained a stable relationship with predicted visitation for both short- and long-distance trips. These are conditional patterns from a fitted observational model and should be interpreted as evidence of modeled distance-attribute tradeoffs, not as causal effects of changing hospital characteristics.

4.4 | Combined Effects of Hospital Ratings and Population SES on Visitation Dynamics

Figure 9 extends the Rating analysis by asking whether the travel-time threshold at which patients begin to favor a more highly rated hospital differs across community socioeconomic contexts in the model simulations. Solid curves represent the higher value of each panel variable and dotted curves represent the lower value; yellow and red indicate the highest and lowest Hospital Rating, respectively. All predictions retain the original observed Drive Times and spatial coordinates. An earlier yellow-red intersection indicates that the higher-rated hospital becomes comparatively favored at a shorter travel time and is therefore interpreted as greater or earlier model-implied attention to Rating.

The results show that this Rating threshold is not uniform across population groups. Communities with a higher percentage of residents aged 65 or older crossed at approximately 12.86 min, well before communities with a lower percentage (18.47 min). This suggests that, once travel is required, older-population communities begin to favor a more highly rated hospital at a substantially shorter travel time. Hispanic and under-18 population scenarios showed the same directional pattern, although with smaller differences (16.40 vs. 19.23 min and 15.85 vs. 19.41 min, respectively).

Asian population composition also showed an earlier Rating response in the higher-value scenario. The higher-Asian solid curves crossed at approximately 12.73 min, compared with 17.86 min for the lower-Asian dotted curves. This result indicates that communities with a higher Asian population share were predicted to begin favoring the more highly rated hospital at a shorter travel time. White population composition showed the opposite relationship: the higher-White scenario crossed at approximately 18.28 min, whereas the lower-White scenario crossed at 13.21 min. Thus, communities with a lower White population share began favoring a highly rated hospital earlier, while Rating differentiated hospital choice less strongly at shorter distances in higher-White communities.

The central result of Figure 9 is not simply that patients consider Rating during longer-distance travel, but that the distance at which Rating begins to differentiate hospital choice varies across community contexts according to the model simulations. Higher-over-65 and higher-Asian communities exhibited notably earlier transitions, and higher-White communities exhibited a later transition. Because these are synthetic area-level scenarios estimated from aggregated CBG data, they describe

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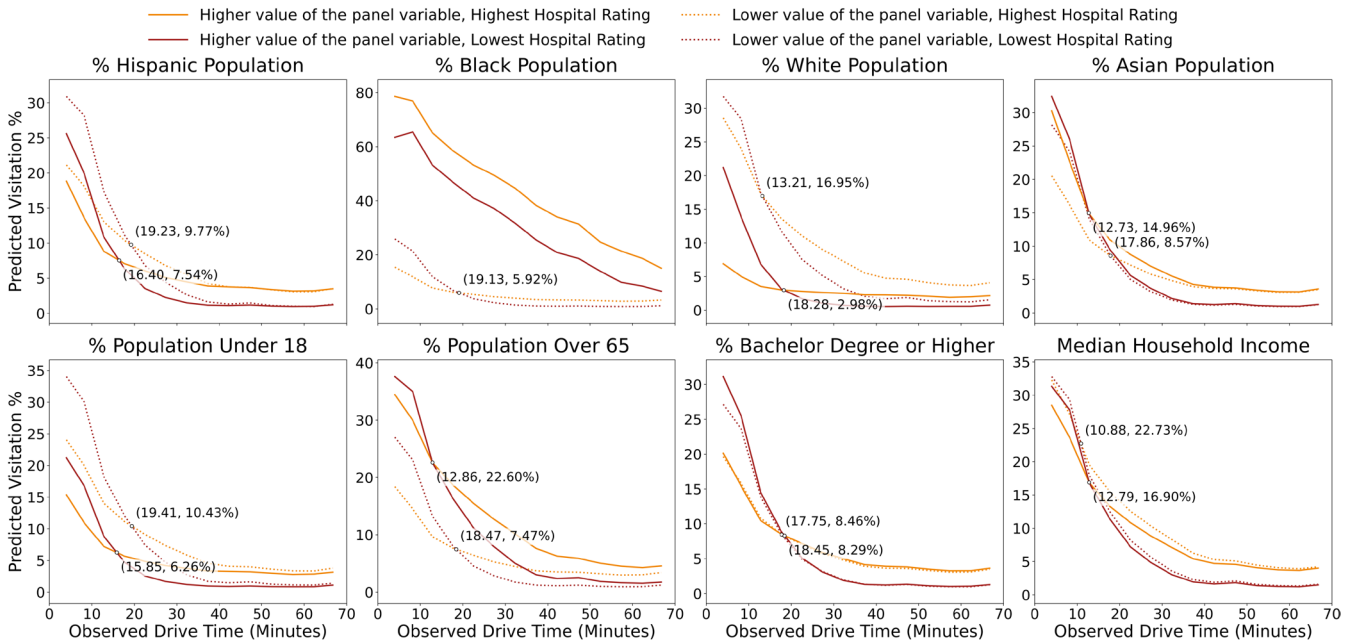


FIGURE 9 | Combined effects of hospital ratings and population SES on visitation distance decay patterns.

model-implied contextual differences and should not be interpreted as individual-level preferences or causal effects.

5 | Discussion

5.1 | Significant Implications

This study shows that hospital visitation is shaped by a distance-dependent tradeoff between local convenience and destination attractiveness. In the model simulations, Drive Time remained the dominant constraint, but the scenario analysis indicates that some hospital attributes become more consequential once patients travel beyond a threshold. In particular, the higher-Rating scenario became favored after approximately 12.78 min, suggesting that perceived hospital attractiveness plays a larger role when patients move beyond local hospital use. In contrast, the stable ordering of the All Bed Occupancy Rate curves indicates a persistent association across both short- and longer-distance visits.

The study also highlights the value of integrating social sensing and large-scale mobility data to examine revealed healthcare travel behavior. Google Maps ratings and reviews capture perceived destination attractiveness, while Advan Research mobility data accessed through Dewey Data enable analysis of observed hospital visitation flows across metropolitan communities. These sources provide a more behavior-oriented view of healthcare mobility and destination choice than conventional accessibility measures alone.

The comparative results demonstrate the value of heterogeneous graph representation. HGNN achieved the best NRMSE, SMAPE, and CPC at all three reported test proportions, while Deep Gravity consistently ranked second. The moderate performance gap indicates that nonlinear origin–destination interactions explain much of the structure, but training-edge message passing and type-specific CBG and hospital representations provide additional predictive information.

The Rating threshold varied across socioeconomic contexts. Communities with higher proportions of residents aged 65 or older and Asian residents began to favor the higher-rated hospital substantially earlier than their corresponding lower-value communities. In contrast, the higher-White scenario crossed later than the lower-White scenario. These contrasts show that the importance attached to hospital Rating during longer-distance travel is heterogeneous rather than uniform across communities. Although the specific effect sizes and distance thresholds reflect the spatial configuration, hospital system, population composition, and mobility-data coverage of Harris County, the broader contribution extends beyond this case. The study provides a transferable framework for modeling realized healthcare destination-choice behavior by representing residential communities and hospitals as interconnected entities and examining how travel burden modifies the relevance of hospital attributes across socioeconomic contexts. This framework can be applied in other metropolitan healthcare systems to evaluate whether similar distance-attribute tradeoffs and community-level heterogeneity emerge under different geographic and institutional conditions.

For transportation and healthcare planning, these findings provide an empirical basis for identifying where travel burden may constrain hospital choice and where hospital attractiveness may encourage patients to travel beyond nearby facilities. The distance thresholds identified for hospital attributes can help planners anticipate how the spatial distribution of hospital capacity, occupancy, and perceived attractiveness may shape visitation demand across service areas. The heterogeneous Rating-Drive Time relationships across community contexts may also help identify populations for whom distance and perceived hospital attractiveness interact differently, supporting more context-sensitive assessments of healthcare access and resource distribution.

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5.2 | Limitations and Future Research

This study includes CBG-level health-insurance coverage from the 2023 ACS, but this measure remains an area-level proxy. It does not identify an individual visitor's insurance status, plan type, provider network, reimbursement rules, or whether a specific hospital was in network. These unobserved financial and institutional constraints may influence hospital choice independently of neighborhood coverage rates. Future research linking privacy-protected patient or survey data with plan-specific network information could better evaluate how insurance interacts with travel burden and hospital attractiveness.

Second, mobility-based visitation data are derived from observed mobile devices and are therefore subject to coverage and representativeness limitations. Individuals without smartphones, those who disable location services, and population groups underrepresented in the underlying device panel may be incompletely captured. If device coverage varies across socioeconomic groups or geographic areas, the estimated visitation flows and socioeconomic relationships may partly reflect differences in panel participation rather than healthcare behavior alone. In addition, a recorded device visit does not necessarily represent a verified patient encounter and may include employees, accompanying persons, or other visitors. The data also lack diagnoses, treatment types, and clinical outcomes. Although normalizing visitation within each CBG improves the comparability of destination-choice patterns, it does not eliminate these sources of selection and measurement bias. The findings should therefore be interpreted as hospital visitation patterns among observed devices rather than population-representative estimates of complete clinical healthcare utilization.

Third, this study relies on 4-year averaged visitation flows, which do not fully capture temporal variations and evolving healthcare mobility patterns. This limitation is particularly important in the context of the COVID-19 pandemic, which substantially altered healthcare-seeking behavior and hospital visitation dynamics (Figure 10). Compared with the COVID-19 crisis period (2020–2021), the recovery phase (2022–2023) exhibited substantial increases in overall visitation intensity, particularly for longer-distance hospital trips. These changes suggest that the pandemic may have altered distance-attractiveness tradeoffs, healthcare-seeking preferences, and long-distance visitation behavior across communities. Future studies could employ longitudinal mobility analyses to investigate how healthcare-related travel behavior evolves under major public health disruptions and recovery processes.

Fourth, socioeconomic variables in this study were measured at the Census Block Group level, meaning that the observed relationships represent area-level associations rather than individual-level behavioral preferences. Therefore, findings related to race, ethnicity, education, and income should not be interpreted as direct evidence of individual decision-making behavior. Future studies integrating individual-level survey or patient data could provide more direct evidence regarding heterogeneous healthcare destination-choice behavior across populations.

Fifth, restricting the study area to Harris County may introduce geographic edge effects. Residents of CBGs near the county boundary may visit hospitals outside Harris County, while patients residing in adjacent counties may visit the hospitals included in this study; these cross-boundary flows are not fully represented. Consequently, normalizing each CBG's observed visits among the 35 included hospitals may overstate the relative allocation to hospitals within Harris County for some boundary CBGs, while the catchment areas of hospitals near the county edge may be truncated. These effects may influence the estimated spatial interaction patterns and distance-attribute relationships, particularly around the study-area periphery. Future research could use a broader multi-county healthcare market or buffer area and conduct sensitivity analyses that exclude boundary CBGs to evaluate the magnitude of this potential edge effect.

Finally, this study focuses exclusively on general acute care hospitals to maintain consistency in healthcare functions and visitation purposes across facilities. However, different healthcare facility types, such as specialty hospitals, rehabilitation centers, urgent care clinics, and chronic-care facilities, may exhibit substantially different healthcare mobility and destination-choice mechanisms. Future studies could extend this framework to multiple healthcare facility categories and compare how

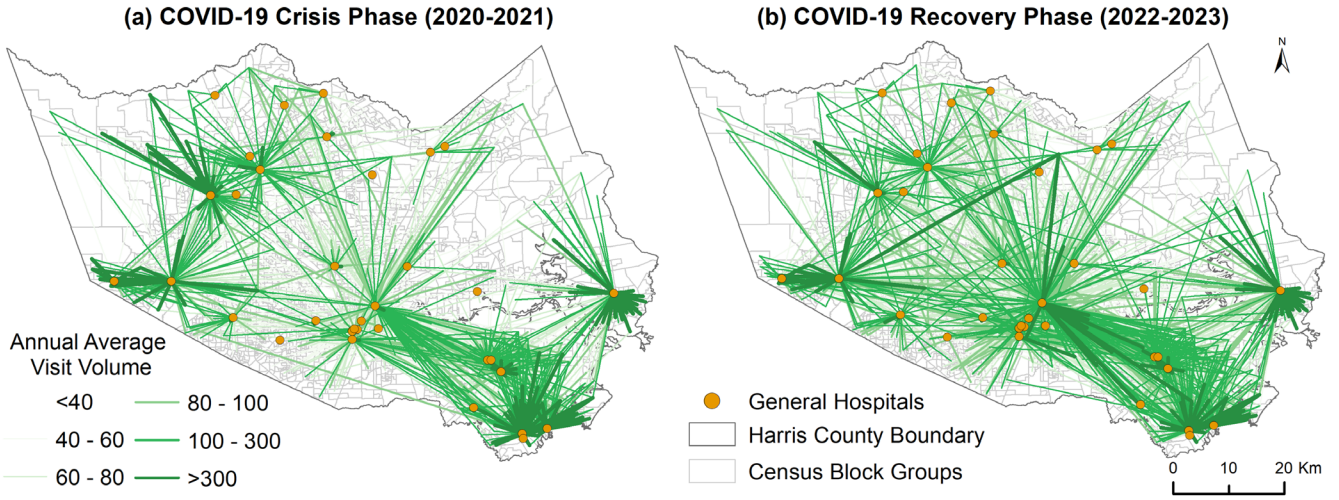


FIGURE 10 | Spatial patterns of annual average visits to general hospitals in Houston during (a) the COVID-19 crisis phase (2020–2021) and (b) COVID-19 recovery phase (2022–2023).

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distance-attractiveness tradeoffs vary across different healthcare systems and visitation contexts.

6 | Conclusion

Hospital visitation reflects a heterogeneous healthcare travel process shaped by travel burden, destination attributes, spatial context, and neighborhood socioeconomic conditions. Using 4 years (2020–2023) of Advan Research mobility data accessed through Dewey Data, hospital attributes, Google Maps indicators, 2023 ACS characteristics including insurance coverage, and observed Drive Times, this study modeled 16,783 observed flows from 1155 CBGs to 35 general acute care hospitals. Drive Time remained the dominant predictor, but HGNN simulations showed that destination attributes influence choice differently over distance. Most notably, the higher-Rating hospital became comparatively favored after approximately 12.78 min, whereas All Bed Occupancy Rate retained a stable relationship with predicted visitation across short- and longer-distance trips.

The Rating-by-SES scenarios further showed that the distance at which Rating begins to differentiate hospital choice varies across community contexts. Higher-over-65 and higher-Asian communities exhibited substantially earlier crossovers than their lower-value counterparts. The higher-White scenario crossed later than the lower-White scenario. These area-level synthetic scenarios identify heterogeneous model-implied sensitivity to Rating during longer-distance travel, but do not establish individual preferences or causal effects.

Among the evaluated models, HGNN achieved the strongest predictive performance across 10%, 20%, and 30% test proportions and all three metrics, with Deep Gravity consistently ranking second. This result indicates that hospital visitation contains both gravity-like origin–destination structure and additional information captured through heterogeneous node representations and training-edge message passing. The findings support aligning model architecture with the relational and behavioral structure of healthcare mobility while maintaining strict separation between training-edge message passing and held-out prediction queries.

Overall, this study contributes to a more behavior-oriented understanding of healthcare-related mobility by conceptualizing hospital visitation as a heterogeneous destination-choice process rather than solely an accessibility outcome. The findings highlight the importance of considering both travel burden and destination attractiveness, together with socioeconomic heterogeneity, when examining healthcare mobility patterns and hospital choice behavior.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The mobility data used in this study were obtained from the Advan Research Foot Traffic/Weekly Patterns Plus dataset through the Dewey Data platform. Due to data licensing restrictions, the original data cannot be redistributed by the authors. Eligible academic researchers may request access through Dewey Data at https://www.deweydata.io/data-partners/advan. The dataset is cited as Advan Research (2025), https://doi.org/10.82551/C103-N851.

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